Type MS · Calculate a percentage change or percentage difference

Answer 1 of 11 · Question 6 part c.ii · 3 marks

2020 JAN · Paper 01 · find part c.ii in this mark scheme extract

Number				
6	(a)	bronchus/bronchi (1)		1
	(b) (i)	• inhale and exhale/breathe in an out (1) • through mouthpiece (1) • floating air chamber rises/falls (1) • pen records a trace on (the rotating) graph paper/drum (1)		4
	(ii)	Do not share the mouthpiece with another person/disinfect mouthpiece before use by another person/after use (1)		1
	(iii)	Any two from: • airways /bronchi/bronchus constricted/obstructed/reduced diameter (1) • less/reduced airflow (1) • per unit time (1)		2
	(c) (i)	cannot be transmitted from one person to another/not contagious/not caused by a pathogen/bacteria/virus/microorganism (1)		1
	(ii)	9.5 - 7 = 2.5(1) 2.5 ÷ 7 x 100(1) 36(%) (1)	Full marks for correct final answer Allow 35.7 for final answer	3
	(iii)	people taking less exercise/ consuming too much fat/sugar/ taking in more energy than is being used/ increase in the number of people smoking/drinking more alcohol/obesity (1)	Allow other valid lifestyle factor	1

Total for Question 6 = 13 marks

Type MS · Calculate a percentage change or percentage difference

Answer 2 of 11 · Question 5 part c.i · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c.i in this mark scheme extract

5(a)	movement of a substance against a concentration gradient/low to high concentration; using energy/ATP;		**(2)**

Question Number	Answer	Additional guidance	Mark
5(b)	• cut pieces of potato to same size/length/mass; • weigh each piece; • place into different solutions; • leave for at least 30 minutes; • remove and dry; • re-weigh/re-measure length; • determine which concentration has no / least change in mass/length;		**(5)**

Question Number	Answer	Additional guidance	Mark
5(c)(i)	0.10 × 100; 7.90 1.3;	Ecf allow correct answer to incorrect figures used in calculation Allow 1.26 / 1.27 for final correct answer	**(2)**

Type MS · Calculate a percentage change or percentage difference

Answer 2 of 11 · Question 5 part c.i · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c.i in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	initial mass/volume/size varies; greater change in mass/volume if potato is bigger; provides a standardised result; allows comparison;		**(2)**
Question Number	Answer	Additional guidance	Mark
5(c)(iii)	• plot a graph; • concentration vs percentage change; • find (from graph) concentration at which percentage change is zero/lowest change; • this is the concentration of the contents;		**(4)**

Total 15 Marks

Type MS · Calculate a percentage change or percentage difference

Answer 2 of 11 · Question 5 part c.i · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c.i in this mark scheme extract

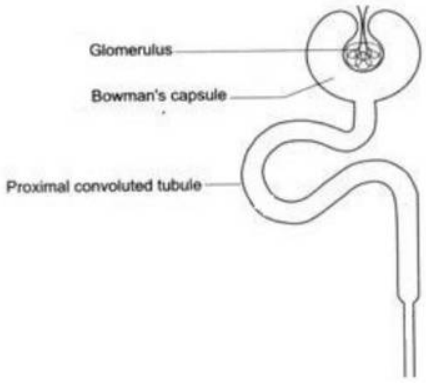
Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Calculate a percentage change or percentage difference

Answer 3 of 11 · Question 9 part a.i · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

9 (a) (i)	• evidence of 23500 – 4000 (= 19500) (1) • 19500 ÷ 4000 (x 100)(1) • 488(%)	Allow full marks for correct final answer. Allow 487.5(%) Allow ecf if incorrect answer from mp1 – max 2 marks	3
(ii)	other lifestyle factors/genetic predisposition could have an influence/named lifestyle factor(1)	Allow gender/age	1
(b)	A diagram similar to:  	One mark for diagram. One mark for each correct label.	4
(c)	any four from • ultrafiltration does not take place(1) • in glomerulus(1) • less/no waste products/named waste product/water/glucose pass through to tubule(1) • greater volume of waste in blood/more water in blood(1) • glucose/ions not reabsorbed(1) • glucose found in urine(1) OR any four from • blood not filtered effectively/ultrafiltration ineffective(1) • in glomerulus (1)		4

Type MS · Calculate a percentage change or percentage difference

Answer 3 of 11 · Question 9 part a.i · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

	• larger molecules/named larger molecules pass through to tubule/Bowmans capsule(1) • not all glucose/ions reabsorbed(1) • glucose/protein found in urine/urine more concentrated(1)		

Total for Question 9 = 12 marks

Type MS · Calculate a percentage change or percentage difference

Answer 4 of 11 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Number				
5 (a)	(i)	• axes labelled (1) • suitable scale (1) • independent variable on X axis (1) • correct plots (2) • suitable curve (1)	-1 for each incorrect plot	6
	(ii)	• mean/average calculated (1) • reliable / reproducible / repeatable (1) • eliminate anomalies (1)		2
(b)	(i)	70 x 100 (1) 90 78%/77.78/77.8 (1)	correct answer = full marks	2
	(ii)	• travels in the blood (1) • time to reach/affect nerve (1)	R heart	2
(c)		• carries impulses to heart muscle (1) • reduces rate of contraction (1)		2

Total for question = 14 marks

Type MS · Calculate a percentage change or percentage difference

Answer 4 of 11 · Question 5 part b.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate a percentage change or percentage difference

Answer 5 of 11 · Question 6 part b.ii · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Calculate a percentage change or percentage difference

Answer 5 of 11 · Question 6 part b.ii · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	

Type MS · Calculate a percentage change or percentage difference

Answer 6 of 11 · Question 4 part c.i · 3 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

Number					
4	(a)	(i)	• eat too much; • too little exercise; • increased fat storage;		3
		(ii)	• heart disease/attacks/strokes; • causes joints in legs to develop arthritis;	Allow high blood pressure/hypertension	2
	(b)	(i)	C (pancreas); A is for excretion B is site of insulin action D produces other hormones		1
		(ii)	B (liver); A is for excretion C is site of production D produces other hormones		1
	(c)	(i)	26378 - 22092 = 4286; $\frac{4286 \times 100}{22092}$; = 19.4%;	Allow $\frac{26378 - 22092}{22092}$ for 2 marks Ecf from first calculation max 2 marks	3
		(ii)	0.9 x 3.8 = 3.4(2) million;		1
	(d)		any three from • less blood flows; • death of tissues; • increased risk of infection; • less white blood cells to deal with infection;		3

Total 14 marks

Type MS · Calculate a percentage change or percentage difference

Answer 6 of 11 · Question 4 part c.i · 3 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate a percentage change or percentage difference

Answer 7 of 11 · Question 3 part b.iii · 3 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

3 (a) (i)	Any five from the following: • place water in the test tube; • take the (starting) temperature of the water; • attach the test tube to a clamp stand; • measure the mass of a piece of food/weigh food; • place food on the mounted needle (under the test tube); • burn the food; (using a Bunsen burner) • take the (final) temperature of the water;		5
(ii)	Any two from the following: • volume/mass of water; • distance of the food from the test tube; • mass of food;	ignore amount ignore amount/size	2
(iii)	Any two from the following: • hold the burning food at arm's length; • reference to sharpness of needle; • goggles; • reference to food allergies/don't taste food;	any ref to being burnt/hot water	2
(b) (i)	Any two from the following: • males need more energy than females; • the amount of energy required decreases with age; • the amount of energy needed by males decreases more (overall) than females;		2
(ii)	**D** - Males use more muscle mass than females;		1
(iii)	• (2629 – 2103) = 526; • 526 ÷ 2103; • (x 100) = 25 (%);	if divide by 2629 and give 20% then two marks Total 15 marks	3

Type MS · Calculate a percentage change or percentage difference

Answer 7 of 11 · Question 3 part b.iii · 3 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Calculate a percentage change or percentage difference

Answer 8 of 11 · Question 7 part c.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Calculate a percentage change or percentage difference

Answer 8 of 11 · Question 7 part c.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Calculate a percentage change or percentage difference

Answer 9 of 11 · Question 5 part b.ii · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

5 (a) (i)	A = eardrum; B = cochlea;		2
(ii)	Any two from the following • collects sound waves/vibrations/determines direction of sound; • directs them through the auditory / ear canal; • to the eardrum;		2
(b) (i)	• axes labels; • units; • bars correctly plotted; • scale;		4
(ii)	• (20 – 14) 6; • (6 ÷20) -0.30; • (x100) -30.00%	Allow ecf is minus symbol not given (Max 2 marks)	3
(c) (i)	Any two from the following • gender variation; • some people may have an ear defect;		2
(ii)	EITHER • check results; • with (data) from another group; OR		2

Type MS · Calculate a percentage change or percentage difference

Answer 9 of 11 · Question 5 part b.ii · 3 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

	• repeat the experiment; • calculate mean;		
			Total 15

Question number	Answer	Notes	Marks

Type MS · Calculate a percentage change or percentage difference

Answer 10 of 11 · Question 8 part b.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

8	(a)	(i)	Any two from • DNA/RNA/genetic material; • surrounded by a protein coat/capsid; • (protein) spikes on outer surface;		2
		(ii)	Any five from • transcription (of DNA); • into mRNA; • translation (of mRNA) at the ribosome; • by tRNA; • reference to (tRNA) anticodon pairing with (mRNA) codon; • amino acids assembled to form (viral) proteins;		5
	(b)	(i)	Human Immunodeficiency virus/HIV;		1
		(ii)	• (1400000 – 650000) = 750000; • (750000 ÷ 1400000) (x 100) = 54%	Allow any figure that rounds to 54	2

Total for Question 8 = 10 Marks

Type MS · Calculate a percentage change or percentage difference

Answer 11 of 11 · Question 3 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

3 (a)	(i)	250:750; 1:3;	full marks for correct answer with no working	2
	(ii)	17 500 - 5600 = 11900; 11 900 × 100; 5 600 = 210%;	full marks for correct answer with no working 213/212.5 = two marks	3
	(iii)	• other parts of the body/organs; • liver/kidney/lungs/arteries/veins; • needs some blood supplied/blood flows to these organs;		3
(b)	(i)	• large/big/huge increase in total flow; • large/big/huge increase in flow skeletal muscles/skin; • decrease for gut;		3
	(ii)	• heat generated(body) temperature increases during exercise; • from respiration; • carried by blood/more blood to skin; • (heat) lost through skin/by radiation;		4

Total 15 marks

Type MS · Calculate a percentage change or percentage difference

Answer 11 of 11 · Question 3 part a.ii · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks